

## sbi\_po\_2025\_mains\_unknown\_shift\_cba8c243

SBI PO Mains 2025

16 questions

The first term of the series which consist of six terms is  $m^\circ$ . The second term of the series is the smallest even prime number. The third term of the series is the cube of the second term. The fourth term of the series is square of third term. The fifth term of the series is 16 times the fourth term of the series. ( $m$  is a natural number)

1. Find the sixth term of the series.

- (a) 1024
- (b) 31768
- (c) 32766
- (d) 32768
- (e) 2244

2. Find the ratio of second term and fifth term of the series.

- (a) 10:21
- (b) 1:8
- (c) 5:324
- (d) 1:512
- (e) 21:44

3. The sixth term is what percentage of 32 time the fifth term of the series.

- (a) 5
- (b) 4
- (c) 3
- (d) 2
- (e) 1

4. If another series starts with 1.5, then find the fourth term of the new series.

- (a) 90
- (b) 93
- (c) 102
- (d) 99
- (e) 96

5. A cuboid has a total surface area (TSA) of 94 sq. cm and a volume of 60 cubic cm. If its length is  $l$  cm, breadth is  $b$  cm, and height is  $h$  cm,  $l$ ,  $b$ , and  $h$ , assuming all are positive and consecutive integers. Given,  $l < b < h$ . Quantity I: Find lateral Surface Area (LSA) of Cuboid. Quantity II: Total Surface Area of a cube with side is  $?b?$ .

- (a) Quantity I > Quantity II
- (b) Quantity I < Quantity II
- (c) Quantity I ? Quantity II
- (d) Quantity I ? Quantity II
- (e) Quantity I = Quantity II or no relation

6. The time taken by A to complete a work is 20 days. B takes  $2x\%$  more days than A to complete the same work, and C takes  $x\%$  more days than B to complete the work. The time taken by C alone to complete the work is equal to the sum of the time taken by A and B alone to complete the work. Find the time taken by all to complete 11 times the work.

- (a) 120
- (b) 130
- (c) 100
- (d) 180
- (e) 200

7. Boat A travels a certain distance in downstream and upstream in 27 hours. Find the time taken by boat B to cover the same distance in downstream and in upstream. The downstream speed of boat A and B is 15 km/hr and 10 km/hr. The upstream speed of boat A and B is 12 km/hr and 9 km/hr.

- (a) 22
- (b) 38
- (c) 10
- (d) 18
- (e) 23

**Read the information carefully and answer the questions given below. There are three Sets A, B, and C, which contain a total of 13 unique integers. Total of ?n? prime numbers is distributed among them. Set A has four numbers. The product of the smallest and largest number in Set A is 26. Set B has five numbers, but only two of them are prime numbers (others are composite). Set C has four numbers. The product of the smallest and largest number in Set C is 23, which is the highest number in all three sets.**

8. If two prime number of set B < 11 and set C contains only one prime number, then find which of the following are the correct set of all the prime numbers?

- (a) 2, 13, 13, 19, 23
- (b) 2, 13, 17, 19, 23
- (c) 2, 5, 7, 13, 19, 17, 23
- (d) 2, 7, 11, 13, 17, 19, 23
- (e) 2, 13, 13, 17, 19, 23

9. If  $n < 8$ , and total prime number in set A > set B, then find the which set have least prime number?

- (a) B
- (b) C
- (c) Can be B and C
- (d) Can't be determined
- (e) None of these

10. If the sum of all elements in set C is 43 and there is total seven prime numbers in all three sets, and the non-prime numbers in set C are consecutive even numbers, then find the sum of prime numbers in set A, 2nd number of set A > 3a

- (a) 21
- (b) 15
- (c) 27
- (d) 30
- (e) 33

11. If set C has highest prime numbers, then find minimum possible value of ?n??

- (a) 8
- (b) 6
- (c) 7
- (d) Can't determined
- (e) None of these

24. A, B, C, D, and E are five persons, and the total weight of all the persons is 200 kg. The total weight of A and B is 56 kg, and the weights of A and B are in the ratio of 3:5, respectively. The weight of C is more than A but less than B. Weight of B is less than D, and weight of E is more than D. (Note: the weights of all persons are integers) Which of the statement/s is/are can be correct. I. Maximum possible weight of C is 34 kg. II. Maximum possible weight of E is 74 kg. III. Maximum possible weight of D is 56 kg.

- (a) All I, II & III
- (b) Both I & III
- (c) Only I
- (d) Both II & III
- (e) Only II

27. The speed of trains A ( $S_a$ ) and B ( $S_b$ ), in km/hr with  $S_a > S_b$ , are the roots of the equation  $x^2 - 130x + 4200 = 0$ . Let  $T_a$  and  $T_b$  be their usual times in hours to cover a certain distance D km. If their speeds are swapped for the same distance D. (i.e. A travels at  $S_b$  and B travels at  $S_a$ ), train A then takes  $(T_a + 2)$  hours and train B takes  $(T_b - 2)$  hours. Find the sum of the original times (in hours),  $T_a$  and  $T_b$ .

- (a) 22

- (b) 21
  - (c) 25
  - (d) 24
  - (e) 26
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28. The volume of a cuboid is 120 cubic meters, and the lateral surface area of the cuboid is 140 square meters. The length, breadth, and height of the cuboid are  $l$ ?,  $b$ ?, and  $h$ ?, respectively, and they are integers. ( $l > b$ ) Quantity I:  $2l$

-b ?  $h/2$  Quantity II:  $3b \times l/h$  Quantity III:  $2b \times l \div h$

- (a) Quantity I < Quantity II < Quantity III
  - (b) Quantity I < Quantity II > Quantity III
  - (c) Quantity I ? Quantity II = Quantity III
  - (d) Quantity I ? Quantity II > Quantity III
  - (e) Quantity I = Quantity < Quantity III
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29. I.  $x^2 - Px + 32 = 0$  (Roots of the equation is 4 and  $4A$ .)  $2(\text{cube of root of } Y) = \text{Square of root of } Y$ . Quantity I: Find the value of  $2a$ .  $Z^2 \times KZ + 990 = 0$  ( $a$  and  $b$  are root of the equation and the value of  $K = 5P+3$ .) Quantity II: Find the value of  $Y+A$ . Quantity III: Find the value of  $(Y+2) \times \text{Highest root of the equation I}$ .

- (a) Quantity I < Quantity II < Quantity III
  - (b) Quantity I < Quantity II > Quantity III
  - (c) Quantity I ? Quantity II = Quantity III
  - (d) Quantity I ? Quantity II > Quantity III
  - (e) Quantity I = Quantity < Quantity III
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30. Statement I: The length of cuboid B is  $3/4$  of the length of cuboid A. The height of cuboid A is 1.2 times that of cuboid B. The difference between the breadth of cuboids A and B is  $Z$ . The breadth of cuboids A and B is in the ratio of 8:3, respectively. Statement II: The area of a cube is  $225z$  square meters, and the side of the cube is the same as the breadth of the cuboid A. Using both statements determined the value of  $Z$ .

- (a) 30
  - (b) 6
  - (c) 9
  - (d) 12
  - (e) 24
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